

Envision Energy

Design Calculation Description for Sri Lanka 11MW/44MWh project

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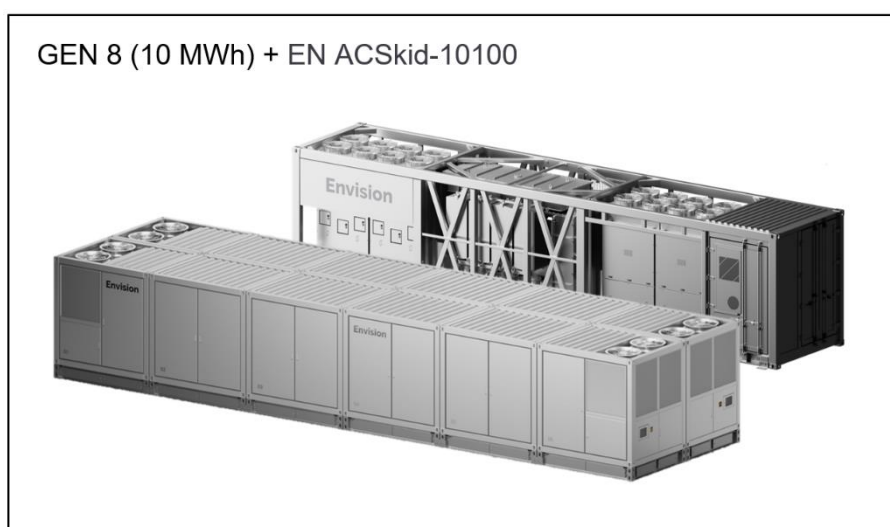
1 Introduction

This document describes design calculations for the Sri Lanka 11MW/44MWh BESS project.

2 Key Project Information

Project name	Sri Lanka 11MW/44MWh BESS project
Environmental conditions	-30°C to +45°C.
Auxiliary losses	Calculated at 35°C
Voltage at Point of Connection (POC)	33kV
Power output at Point of Connection (POC)	11 MW (nominal)
Usable energy at Beginning of Life (BoL)	43.9 MWh offered
Cycles per day	1.1
Proposed solution	2 sets ENS-D06G-24120-10100-000

3 General BESS Design Parameters



AC skid	PCS rated power	4 x 2520 kW
	Transformer voltages	0.69 kV / 33 kV
	Transformer type	3 windings
	Transformer rated capacity	10100 kVA
DC container	Voltage range of DC system	1165 VDC to 1500 VDC
	Rack configuration	10 racks per DC container

4 Proposed BESS Design

4.1 Proposed Overall Solution

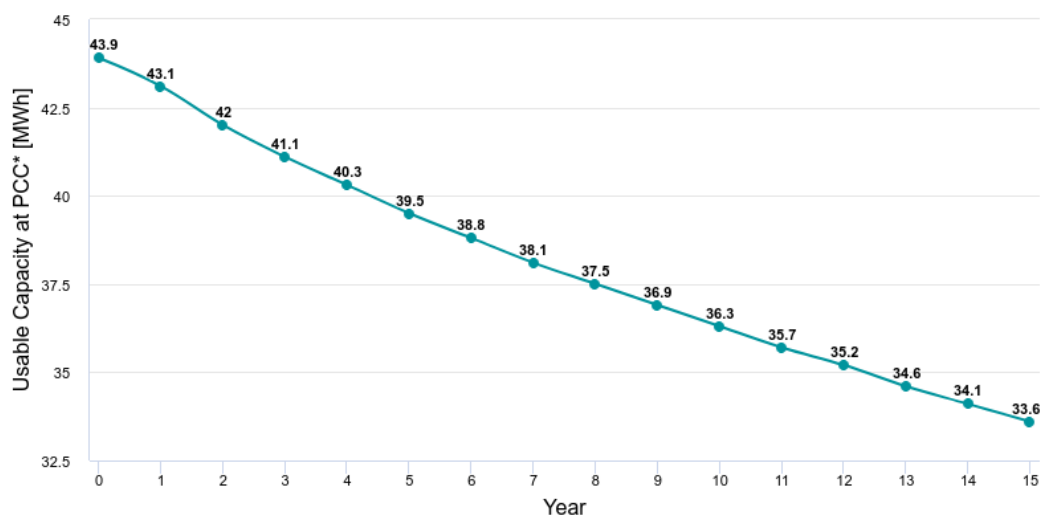
2 sets ENS-D06G-24120-10100-000

Item	Value	Unit	Comments
Rated power	20.1	MVA	
Required active power	11	MW	At PoC
Usable active power	18.7	MW	At PoC
Installed capacity	48.2	MWh	
Usable capacity @ POC	43.9	MWh	At BoL

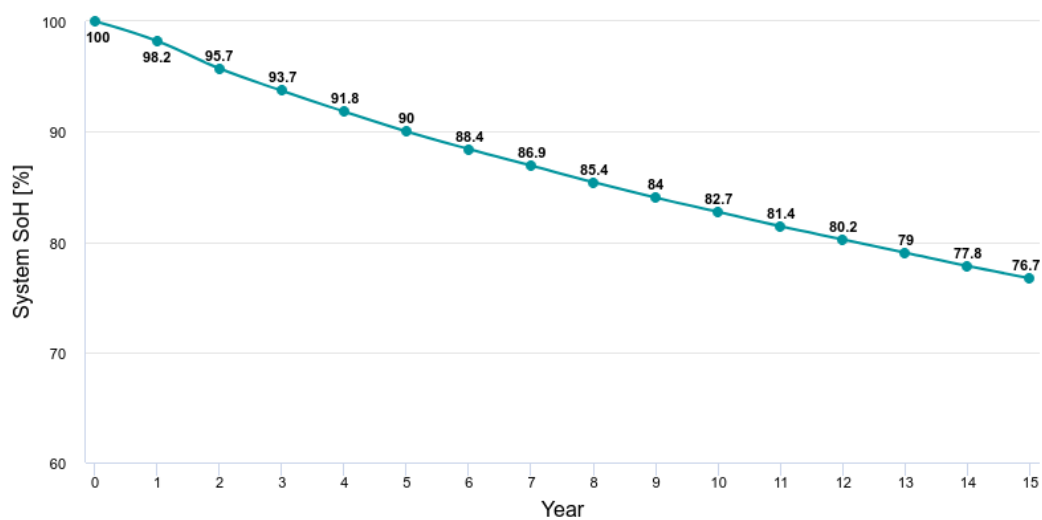
4.2 Performance Curves

1.1 cycles/day										
Year	Usable capacity at 33 kV including aux power (MWh)	Usable capacity at 33 kV excluding aux power (MWh)	System SoH including aux power	System SoH excluding aux power	RTE including aux power	RTE excluding aux power	DC Usable Discharge Capacity (MWh)	DC-DC RTE	Export in Previous Year (MWh)	Cumulative Export (MWh)
BoL	43.9	44.6	100%	100%	86.9%	89.9%	45.8	94.8%		
1	43.1	43.9	98.2%	98.3%	86.6%	89.8%	45	94.7%	17465	17465
2	42	42.8	95.7%	95.9%	86.4%	89.7%	43.9	94.6%	17102	34568
3	41.1	41.9	93.7%	93.8%	86.2%	89.6%	43	94.5%	16703	51271
4	40.3	41.1	91.8%	92%	86.1%	89.5%	42.1	94.4%	16354	67626
5	39.5	40.3	90%	90.3%	86%	89.5%	41.3	94.3%	16036	83662
6	38.8	39.6	88.4%	88.7%	85.8%	89.4%	40.6	94.2%	15740	99403
7	38.1	38.9	86.9%	87.2%	85.7%	89.3%	39.9	94.1%	15462	114865
8	37.5	38.3	85.4%	85.7%	85.6%	89.2%	39.2	94.1%	15198	130064
9	36.9	37.6	84%	84.3%	85.5%	89.2%	38.6	94%	14946	145010
10	36.3	37.1	82.7%	83%	85.4%	89.1%	38	93.9%	14705	159716
11	35.7	36.5	81.4%	81.7%	85.3%	89.1%	37.4	93.9%	14473	174190

12	35.2	35.9	80.2%	80.5%	85.2%	89%	36.9	93.8%	14250	188440
13	34.6	35.4	79%	79.3%	85.1%	88.9%	36.3	93.8%	14034	202475
14	34.1	34.9	77.8%	78.2%	85%	88.9%	35.8	93.7%	13825	216300
15	33.6	34.4	76.7%	77%	85%	88.8%	35.3	93.6%	13623	229924



*Usable capacity at PCC including the auxiliary losses



*Note that the above values are contingent upon the assumed loss factors shown in the following table. Furthermore, it is important to note that these values may exhibit a slight variation of 1-2 % in response to actual site conditions, attributable to fluctuations in ambient temperature (± 5 °C deviation) and the measurement accuracy of external metering equipment (such as PT, CT, etc.). Rest assured, we remain committed to ensuring precision and transparency throughout this project's execution. Furthermore, if the RTE is tested using an electricity meter, we consider that the meter has an error of around 0.5%; therefore, the RTE is deemed compliant within this margin of error.

4.3 Design Sizing

4.3.1 DC Unit

ENS-Do6G-24120-10100-000	Value	Unit	Comments
DC container configuration	6030	kWh	6 racks connected in parallel per Container
Nameplate capacity type 1	24.121	MWh	4 containers connected per AC twin-skid

4.3.2 AC Skid

Item	Value	Unit	Comments
Step up transformer	10.1	MVA	
PCS	2.52	MVA	
AC unit	10.08	MW	MV AC container. 4x PCS & 1x transformer

4.4 Assumptions

Item	Efficiency	Remarks
Calendar degradation	97 %	From cell manufacture to Site Acceptance Test (SAT). Assuming 6 months from Factory Acceptance Test (FAT) to SAT
Usable DC capacity ratio	98 %	Includes battery cell efficiency & DOD limitations
LV DC cables	99.9 %	Assuming that the cables are aluminium @ 30m, provided by BoP
PCS	98.5%	At Rated Power, provided by Envision
LV/MV transformer	99.2%	Compliance with EU Ecodesign standards, provided by Envision
MV/HV transformer	100 %	Assumed, provided by BoP
MV cable	99.6 %	Assumed, provided by BoP
HV cable	100 %	Assumed, provided by BoP

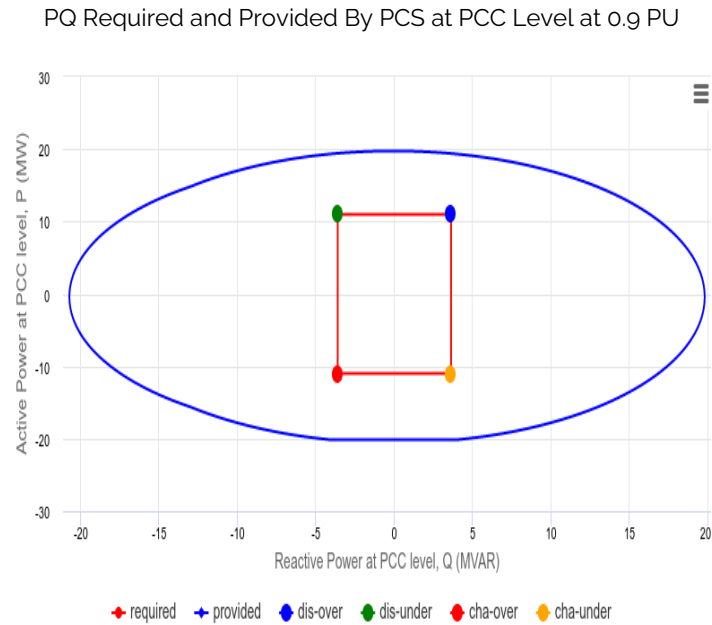
Item	Energy consumption	Remarks
Aux consumption at BoL	0.17 MW x 4 h	Auxiliary consumption cannot be guaranteed separately

Item	Impedance	Remarks
LV/MV transformer	9 %	Compliance with EU Ecodesign standards
MV/HV transformer	16 %	Assumed, provided by BoP

5 PCC Requirements

Parameter	Unit	Discharge-Overexcitation	Discharge-Underexcitation	Charge-Overexcitation	Charge-Underexcitation
P required at PCC	MV	11	11	-11	-11
S required at PCC	MVA	11.58	11.58	11.58	11.58
Q required at PCC	MVAR	3.62	-3.62	3.62	-3.62
Power factor at PCC	-	0.95	-0.95	0.95	-0.95
P required at PCS	MW	11.37	11.36	-10.64	-10.64
S required at PCS	MVA	12.37	11.61	11.68	10.91

Q required at PCS	MVAR	4.87	-2.37	4.82	-2.42
Power factor at PCS	-	0.92	-0.98	0.91	-0.98



*Note that the above power flow calculation results are only valid when the PCS terminal voltage is within 0.9–1.1 pu range.

6 Conclusion

This document describes and proposes 2 sets ENS-Do6G-24120-10100-000 for the Sri Lanka 11MW/44MWh BESS project. It outlines the essential parameters of the DC unit of the BESS with representative images of the DC containers, including details of the performance curves, State of Health (SoH) of battery and Round-Trip Efficiency (RTE) of the battery in accordance with the project's stipulated requirement. The yearly usable energy capacity of the system has also been presented in the performance curve table throughout the design life of the project.

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